Physical Parameters Joint Estimation of Satellite Parabolic Antenna With Key Frame Pol-ISAR Images

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***Abstract*— Physical parameter estimation of satellite is crucial in space situation awareness as it reflects valuable information. Polarimetric inverse synthetic aperture radar (Pol-ISAR) is a powerful sensor for space surveillance, providing rich information for satellite physical parameter estimation. Parabolic antennas, which are widely loaded in remote sensing and communication satellites, have received great attention recently. Dedicated to the space situation awareness issue using Pol-ISAR, a physical parameter joint estimation method of satellite parabolic antenna with key frame Pol-ISAR images is developed in this work. The core idea is to utilize the mapping relationship between parabolic antenna in 3-D space and its projection ellipse in 2-D ISAR image. Under special observation geometry, the closed-form expressions of parabolic antenna physical parameters are deduced for the first time, providing an efficient way for parameter estimation. Moreover, the abundant information within Pol-ISAR images is mined and utilized. Various polarimetric features are adopted for ellipse extraction and the subsequent physical parameter estimation. Compared with single-polarization channel data, the superiorities of polarimetric feature are validated using electromagnetic simulation data.**

***Index Terms*— Joint estimation, parabolic antenna, physical parameter, polarimetric features, polarimetric inverse synthetic aperture radar (Pol-ISAR).**

I. Introduction

NOWADAYS, space situation awareness receives great attention with the utilization of outer space [1], [2]. With the development of inverse synthetic aperture radar (ISAR) system and imaging algorithm [3], the qualities of ISAR images have been improved significantly, providing rich information for space situation awareness.

Physical parameters of satellite, including the size, attitude, and position parameters, which can reflect the movement intentions, payload indexes, and imaging areas, are of vital importance in space situation awareness. Until now, the parameter estimation approaches for space targets can be divided into three parts: the template matching-based method [4], [5], [6], [7], the 3-D reconstruction method [8], [9], [10], [11], [12], and the projection theory-based method [1], [13], [14], [15].

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Template matching-based methods are mainly adopted for cooperative targets, the computer aided design (CAD) models of which are known as a prior or where the supportive devices, such as corner cube reflector (CCR) and light-emitting diode, are equipped. Based on the CAD models of space targets, full-angle datasets of ISAR images are constructed under different observation geometries [4]. The attitude of space target is determined by matching the measured ISAR images with the simulated dataset [4]. Apart from ISAR images, the high-resolution range profile (HRRP) is also adopted for space target attitude estimation [5]. In this vein, a 3-D electromagnetic-model-based algorithm is proposed in [5] for absolute attitude measurement using wideband radar, which can be applied to aircraft, satellite, missiles, and so on. Moreover, if supportive equipment, such as CCR, is available on the satellite, the corresponding attitude information of cooperative targets can be obtained [6]. The track information of Environmental Satellite (EnviSat) from 2013 to 2015 is recorded with global satellite laser ranging stations. By matching this historical database, the current spin parameters of satellite are estimated based on the relative position of CCR [6]. Recently, a satellite attitude inversion method is developed in [7], which utilizes the observation images of satellite and the known satellite position. The proposed method works well for imaging satellite with the help of correct map projection [7]. Nonetheless, all the template-based methods require a well-established dataset and may fail for uncooperative targets. In this vein, more robust parameter estimation method needs to be explored for space targets.

Another parameter estimation approach is developed based on the 3-D reconstruction of space targets. Generally, these kinds of methods involve observing the targets from multiple angles, which can be achieved through long-term ISAR observation [8], [9], [10], [11], [12]. The factorization method, transferred from computer vision, is popularly adopted for 3-D reconstruction of man-made targets [16]. However, accurate association of scatter points can be challenging in practice due to the angular glint phenomenon of space targets [17]. To address this issue, a modified method has been proposed to retrieve the complete scatter's trajectory matrix [8]. Apart from single station system, the interferometric ISAR (InISAR) system, which makes use of spatial information provided by multiple sensors, performs quite well in 3-D reconstruction [9], [10], [11]. It should be noted that the coherent processing

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of multistation InISAR system is still challenging. Recently, the polarimetric ISAR system has gained much attention since the acquired full-polarimetric information is naturally suitable for satellite characteristic interpretation. The full-polarimetric CLEAN approach is successfully adopted in [9] to extract the dominant scattering centers from ISAR images. In addition, combining interferometric phase and 2-D ISAR imaging theory, a 3-D polarimetric InISAR imaging method is developed in [12], which performs well in point cloud generation.

Nowadays, projection theory-based approaches have received much attention [1], [13], [14], [15]. The high-quality ISAR images pave the way for parameter estimation. Projection matrix connects the 3-D space target with 2-D ISAR imaging results, enabling the derivation of 3-D target parameters from sequential ISAR images [1], [13], [14], [15]. In this vein, typical structures, such as linear structure and circle shape structure, are paid special attention to.

For linear structure extraction, background segmentation and morphology techniques are combined in [1], and the attitude information of space target can be derived by parallelogram descriptor and optimization method. An erode structure is adopted in [13] to extract the axis attitude information of space targets, which mainly focuses on the linear structure. However, the abovementioned methods are conducted on binary images. The complex thresholding procedure limits their applications in practice. To this end, peak accumulation values of rotated ISAR images are adopted for linear structure extraction, which can be directly applied on original ISAR images rather than binary images [14]. Apart from model-based approaches, the pix2pix generative adversarial network is developed in [15] for image segmentation and linear structure extraction. For parabolic antenna in circle shape, its attitude information can be extracted using ellipse parameters and optimization algorithms [18]. Therefore, the accurate detection of ellipse structure in ISAR images is quite important. Randomized Hough transform (RHT)-based method utilizes the parametric representation of curve structures with geometric properties of ellipse, such as collinearity and symmetry [19], [20]. However, the time-consuming of RHT-based methods can be quite long, limiting their applications in practice. Apart from RHT-based methods, another technical routine for ellipse detection is edge following [21]. Nonetheless, the edge following methods rely on the accuracy extraction of edge contour, which is a tough task in ISAR images.

The abovementioned methods are developed based on optical images or single-channel ISAR images. In this vein, the abundant information in polarimetric ISAR (Pol-ISAR) data is rarely discovered. For polarimetric information interpretation, a number of advanced polarimetric decomposition methods have been developed over several decades, which can be divided into two categories: incoherent target decomposition [22], [23] and coherent target decomposition [24]. The decomposition results contain different scattering mechanisms, which have potential in edge information reservation. Besides, the superiority of polarimetric information has been demonstrated in structure recognition [25], [26], [27], target detection [28], [29], and ISAR image super-resolution [30]. The potential of polarimetric information in space target attitude estimation has been rarely revealed, which is worthy of paying more attention to. In summary, the parameter estimation of space target is still challenging in the following two aspects. On the one hand, the time-consuming of RHT-based method can be quite long, limiting its application in practice. On the other hand, the polarimetric information in ISAR images, which has potential in promoting the parameter estimation performance, is rarely considered.

To mitigate the abovementioned issues, this work focuses on parabolic antenna component and develops a joint physical parameter estimation method with key frame Pol-ISAR images. The estimated physical parameters include position, size, and attitude parameters. The main contributions of this work fall on the following three parts.

1) To the best of our knowledge, the direct mapping relationships between satellite parabolic antenna parameters in 3-D space and ellipse structure parameters in 2-D ISAR images are revealed for the first time. A novel physical parameter joint estimation method for parabolic antenna is proposed thereafter, the effectiveness of which is validated by point target simulation data.

2) Based on the direct mapping relationships, three key frames are sufficient for physical parameter estimation in extreme case. The closed-form expressions of parameter estimation under specific observation geometries are deduced, providing an efficient way for physical parameter estimation.

3) Various polarimetric features are adopted for physical parameter estimation. Compared with single-polarization channel data, the selected polarimetric decomposition features have superior performance in edge reservation of ellipses and physical parameter estimation of parabolic antennas.

The remainder of this article is organized as follows. Section II concentrates on the development of physical parameter joint estimation method for parabolic antenna. Section III validates the effectiveness of the proposed method using point target simulation data. Experimental studies with electromagnetic simulation data are conducted in Section IV. Some discussions are given in Section V. Conclusions are presented in Section VI.

II. Physicals Parameter Joint Estimation Method

A. Projection Relationship

The edge contour of satellite parabolic antenna in 3-D space is usually in circle shape, while its projection in ISAR image is ellipse structure. The projection illustration is shown in Fig. 1.

The line-of-sight (LOS) vector connects $\boldsymbol{O}_{\text{Target}}$ and $\boldsymbol{O}_{\text{Radar}}$. The red points indicate the edge points of parabolic antenna in 3-D space, while the blue ellipse represents the projection result in 2-D ISAR imaging plane.

The radar LOS is expressed as

$$\mathbf{LOS} = \begin{bmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi & \sin\theta \end{bmatrix}^{\mathrm{T}} \tag{1}$$

where $\theta \in [-90°, 90°)$ and $\phi \in [-180°, 180°)$ are the elevation angle and azimuth angle, respectively. The subscript ^{T} means transposition.

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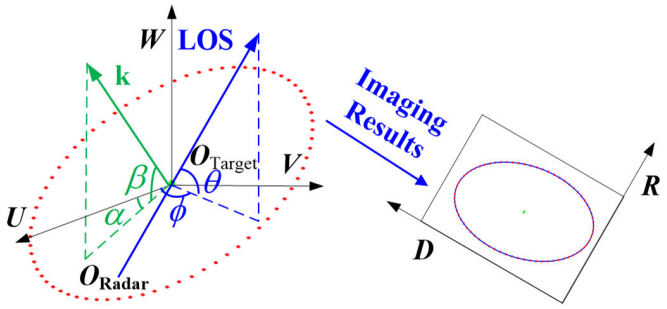


Fig. 1. Geometry projection illustration of ISAR imaging.

The attitude of parabolic antenna is defined by the normal vector of antenna, which is denoted as

$$\mathbf{k} = \begin{bmatrix} \cos\beta\cos\alpha & \cos\beta\sin\alpha & \sin\beta \end{bmatrix}^{\mathrm{T}} \tag{2}$$

where $\beta \in [-90°, 90°)$ and $\alpha \in [-180°, 180°)$ are the pitch angle and yaw angle, respectively.

The $\boldsymbol{R}$-axis of imaging plane is defined by the LOS vector and the $\boldsymbol{D}$-axis is defined by the derivation of LOS vector [1]. The imaging results of parabolic antennas in 3-D space are ellipses in 2-D ISAR images, as shown in the right part of Fig. 1. For edge point $\mathbf{M} = [\,u\ v\ w\,]^{\mathrm{T}}$ in 3-D space, its projection result in 2-D $\boldsymbol{R}-\boldsymbol{D}$ imaging plane is denoted by $\mathbf{N} = [\,r\ d\,]^{\mathrm{T}}$. The projection relationship is

$$\mathbf{N} = \mathbf{P}\cdot\mathbf{M} \tag{3}$$

and $\mathbf{P}$ is the projection matrix, denoted as

$$\mathbf{P} = \begin{bmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi & \sin\theta \\ P_{21} & P_{22} & \cos\theta\theta' \end{bmatrix} \tag{4}$$

where P_{21} and P_{22} are defined as

$$\begin{cases} P_{21} = -\sin\theta\cos\phi\theta' - \cos\theta\sin\phi\phi' \\ P_{22} = -\sin\theta\sin\phi\theta' + \cos\theta\cos\phi\phi' \end{cases} \tag{5}$$

where θ' and ϕ' are the instantaneous velocities of elevation and azimuth angles, respectively.

Moreover, if the scatter point $\mathbf{M}$ is rotated along the $\boldsymbol{O}_{\mathrm{Target}}$-$\boldsymbol{UVW}$ coordinate with angle $[\,\theta_\mu\ \theta_v\ \theta_\omega\,]^{\mathrm{T}}$, the rotated point $\hat{\mathbf{M}}$ is

$$\hat{\mathbf{M}} = \mathbf{R}_W\cdot\mathbf{R}_V\cdot\mathbf{R}_U\cdot\mathbf{M} \tag{6}$$

where $\mathbf{R}_U$, $\mathbf{R}_V$, and $\mathbf{R}_W$ are the rotation matrices along the $\boldsymbol{U}$-, $\boldsymbol{V}$-, and $\boldsymbol{W}$-axes, respectively. In detail,

$$\mathbf{R}_U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_\mu & -\sin\theta_\mu \\ 0 & \sin\theta_\mu & \cos\theta_\mu \end{bmatrix} \tag{7}$$

$$\mathbf{R}_V = \begin{bmatrix} \cos\theta_v & 0 & -\sin\theta_v \\ 0 & 1 & 0 \\ \sin\theta_v & 0 & \cos\theta_v \end{bmatrix} \tag{8}$$

$$\mathbf{R}_W = \begin{bmatrix} \cos\theta_\omega & -\sin\theta_\omega & 0 \\ \sin\theta_\omega & \cos\theta_\omega & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{9}$$

Under this condition, the projection relationship is

$$\mathbf{N} = \mathbf{P}\cdot\mathbf{R}_W\cdot\mathbf{R}_V\cdot\mathbf{R}_U\cdot\mathbf{M} = \mathbf{P}'\cdot\mathbf{M} \tag{10}$$

where $\mathbf{P}' = \mathbf{P}\cdot\mathbf{R}_W\cdot\mathbf{R}_V\cdot\mathbf{R}_U$. It should be noted that (6)–(10) contain the most complex situations of ISAR imaging. In practice, chances are that only one rotational axis is considered for the sake of high imaging quality.

B. Ellipse Parameters

The projection results of parabolic antennas in ISAR images are ellipses in the general case, which can be characterized as

$$\frac{\left[(x-x_0)\cos\varphi - (y-y_0)\sin\varphi\right]^2}{R_1^2} + \frac{\left[(x-x_0)\sin\varphi + (y-y_0)\cos\varphi\right]^2}{R_2^2} = 1 \tag{11}$$

where x_0 and y_0 are the coordinates of ellipse center, R_1 and R_2 are the intercepts on the $\boldsymbol{D}$- and $\boldsymbol{R}$-axes, and φ is the inclination angle.

Equation (11) can be rewritten as a linear equation as

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + 1 = 0. \tag{12}$$

To solve the five parameters in (12), $n(n \geq 5)$ points and the least squares method are adopted in this work.

Point target extraction is the key process of solving linear equations, the details of which are as follows. First, pixel with the highest value is chosen. Second, a guard window (e.g., 5×5) is defined around the selected pixel, and the neighboring pixels are set to zero. Then, pixel with the next highest feature value is iteratively chosen until n ($n = 7$ in this work) points are collected. Finally, n points are used to solve the linear equations using the least squares method.

Based on the extracted strong points, parameters in (12) can be obtained. The ellipse parameters in (11) can be transformed from the parameters in (12) as

$$\begin{cases} x_0 = \dfrac{BE - 2CD}{4AC - B^2} \\ y_0 = \dfrac{BD - 2AE}{4AC - B^2} \\ R_1 = \sqrt{\dfrac{2\left(Ax_0^2 + Cy_0^2 + Bx_0y_0 - 1\right)}{A + C - \sqrt{(A-C)^2 + B^2}}} \\ R_2 = \sqrt{\dfrac{2\left(Ax_0^2 + Cy_0^2 + Bx_0y_0 - 1\right)}{A + C + \sqrt{(A-C)^2 + B^2}}} \\ \varphi = \dfrac{1}{2}\arctan\left(\dfrac{B}{A-C}\right). \end{cases} \tag{13}$$

In practice, the accurate extraction of ellipse structure is the basis for joint parameter estimation. In this work, theoretical analysis and experimental studies are conducted on Pol-ISAR images with satisfying imaging quality [e.g., high signal-to-noise ratio (SNR)] and relative salient ellipse structures.

C. Polarimetric Feature Extraction

Scattering points can be selected from various feature maps. Considering the rich information of Pol-ISAR data, various polarimetric features can be extracted and adopted for scattering point extraction.

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Taking Pol-ISAR data in horizontal and vertical polarization base for instance, the fully polarimetric data are

$$\mathbf{S} = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{14}$$

where S_{HV} denotes the backscattered coefficient from vertical transmitting and horizontal receiving polarization. Other three terms are similarly defined. Note that the reciprocal condition is satisfied in general case, where $S_{\mathrm{HV}} = S_{\mathrm{VH}}$.

Total backscattering power, which is also noted as SPAN, is defined as

$$\mathrm{SPAN} = |S_{\mathrm{HH}}|^2 + 2|S_{\mathrm{HV}}|^2 + |S_{\mathrm{VV}}|^2 \tag{15}$$

where $|\cdot|$ denotes the complex magnitude.

The Pauli vector is defined as

$$\mathbf{k}_{\mathrm{P}} = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\mathrm{HH}} + S_{\mathrm{VV}} & S_{\mathrm{HH}} - S_{\mathrm{VV}} & 2S_{\mathrm{HV}} \end{bmatrix}^{\mathrm{T}}. \tag{16}$$

The polarimetric coherency matrix is defined as

$$\mathbf{T} = \left\langle \mathbf{k}_{\mathrm{P}} \mathbf{k}_{\mathrm{P}}^{\mathrm{H}} \right\rangle \tag{17}$$

where subscript ^{H} denotes the conjugate transpose and $\langle\cdot\rangle$ is the sample average.

Polarimetric target decomposition is a powerful tool for polarimetric data interpretation, which contains two categories: polarimetric incoherent and coherent decompositions. In this work, both decomposition schemes are considered.

Yamaguchi decomposition [22] is a classical incoherent target decomposition method, which divides the polarimetric coherency matrix into four components, including surface scattering, double-bounce scattering, volume scattering, and helix scattering. The decomposition process can be written as

$$\mathbf{T} = f_s \mathbf{T}_{\mathrm{surface}} + f_d \mathbf{T}_{\mathrm{double}} + f_v \mathbf{T}_{\mathrm{volume}} + f_c \mathbf{T}_{\mathrm{helix}} \tag{18}$$

where f_s, f_d, f_v, and f_c are the decomposition features and $\mathbf{T}_{\mathrm{surface}}$, $\mathbf{T}_{\mathrm{double}}$, $\mathbf{T}_{\mathrm{volume}}$, and $\mathbf{T}_{\mathrm{helix}}$ are the corresponding expansion matrices of different scattering mechanisms.

General polarimetric model-based decomposition [23] makes use of all the elements of polarimetric coherency matrix, which can avoid traditional problems such as manual intervention, branch conditions, and negative power. Similarly, the derived four scattering components are odd-bounce scattering, volume scattering, double-bounce scattering, and helix scattering. The general decomposition framework is

$$\begin{aligned} \mathbf{T} = {} & p_s \mathbf{T}_{\mathrm{odd}}(\theta_{\mathrm{odd}}) + p_v \mathbf{T}_{\mathrm{volume}} \\ & + p_d \mathbf{T}_{\mathrm{double}}(\theta_{\mathrm{double}}) + p_c \mathbf{T}_{\mathrm{helix}} + \mathbf{T}_{\mathrm{residual}} \end{aligned} \tag{19}$$

where p_s, p_v, p_d, and p_c are decomposition features; $\mathbf{T}_{\mathrm{odd}}(\theta_{\mathrm{odd}})$, $\mathbf{T}_{\mathrm{volume}}$, $\mathbf{T}_{\mathrm{double}}(\theta_{\mathrm{double}})$, and $\mathbf{T}_{\mathrm{helix}}$ are expansion matrices of different scattering mechanisms; and $\mathbf{T}_{\mathrm{residual}}$ is the residual item in the model inversion procedure.

Krogager decomposition [24] is conducted on polarimetric scattering matrix, which belongs to the coherent decomposition scheme. Three scattering components, named sphere scattering, rotated dihedral scattering, and helix scattering, are derived from the Krogager decomposition as

$$\mathbf{S}_{(\mathrm{H,V})} = e^{\mathrm{j}\phi} \left(e^{\mathrm{j}\phi_{\mathrm{s}}} k_{\mathrm{s}} \mathbf{S}_{\mathrm{sphere}} + k_{\mathrm{d}} \mathbf{S}_{\mathrm{dihedral}(\theta)} + k_{\mathrm{h}} \mathbf{S}_{\mathrm{helix}(\theta)} \right) \tag{20}$$

where k_{s}, k_{d}, and k_{h} are the decomposition features; $\mathbf{S}_{\mathrm{sphere}}$, $\mathbf{S}_{\mathrm{dihedral}(\theta)}$, and $\mathbf{S}_{\mathrm{helix}(\theta)}$ are corresponding matrices of different scattering mechanisms; θ in (20) denotes the orientation angle; and ϕ and ϕ_{s} in (20) mean the absolute and relative phases of polarimetric scattering matrix, respectively.

In this work, three single independent polarization data HH, HV, and VV (HV denotes the polarization channel data of vertical transmitting and horizontal receiving and other items are similarly defined); SPAN feature; four Yamaguchi decomposition features f_s, f_d, f_v, and f_c; four general polarimetric decomposition features p_s, p_v, p_d, and p_c; and three Krogager decomposition features k_s, k_d, and k_h are selected for ellipse structure extraction and the subsequent physical parameter estimation.

D. Joint Parameter Estimation Method

For a certain parabolic antenna, its location in 3-D space can be defined by the center point $\boldsymbol{O}_{\mathrm{Antenna}} = [\mu \quad \nu \quad \omega]^{\mathrm{T}}$, while its size is defined by radius r. Moreover, its attitude is defined by the normal vector $\mathbf{k}$ in Fig. 1, which is characterized by two parameters of yaw angle α and pitch angle β.

In summary, the position, size, as well as attitude parameters form the parameter set $\ddot{\theta}$ as

$$\ddot{\theta} = \begin{bmatrix} \mu & \nu & \omega & r & \alpha & \beta \end{bmatrix}. \tag{21}$$

The details for physical parameter estimation are as follows.

1) Yaw Angle Estimation: First, the yaw angle α is investigated based on the inclination angles of ellipses. Considering the azimuth angle of LOS and yaw angle of target attitude, the equivalent azimuth angle is

$$\psi = \phi - \alpha. \tag{22}$$

When the elevation angle is $\theta = 0°$, the ISAR imaging results are the orthographic projection of targets on imaging plane. Under this condition, the inclination angle φ of ellipse structure is equal to the equivalent azimuth angle ψ. In radar tracking system, the azimuth angle of LOS is known as a prior. Therefore, the yaw angle can be obtained based on the extracted inclination angle.

For F_1 key frame ISAR images with $\theta = 0°$, the yaw angle α can be obtained as

$$\alpha = \frac{1}{F_1} \sum_{f_1=1}^{F_1} \left(\phi_{f_1} - \psi_{f_1} \right) = \frac{1}{F_1} \sum_{f_1=1}^{F_1} \left(\phi_{f_1} - \varphi_{f_1} \right). \tag{23}$$

In practice, one key frame with $\theta = 0°$ is sufficient to derive yaw angle from (23). More key frames help to promote the estimation performance.

2) Radius Parameter Estimation: Second, the size parameter r is investigated. According to the ISAR imaging geometry, when the equivalent azimuth angle is set as 0° (i.e., $\phi = \alpha$), the projection results of parabolic antenna are standard ellipses. In this vein, the scaling effects in range and azimuth directions can be utilized for physical parameter estimation. Since the yaw angle has been estimated according to (23), specific frame ISAR images can be selected for radius parameter estimation, whose azimuth angles are supposed to be $\phi = \alpha$.

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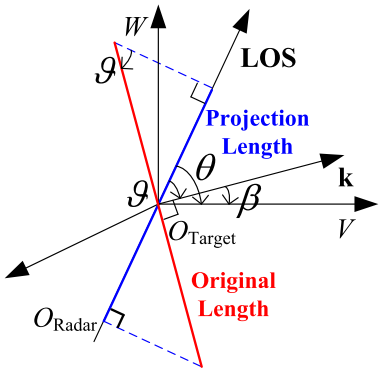


Fig. 2. Scaling effect of ISAR projection geometry.

For simplicity, both the LOS vector and attitude vector **k** are assumed to be in the $\boldsymbol{W}\boldsymbol{O}_{\text{Target}}\boldsymbol{V}$ plane, as shown in Fig. 2. Under this condition, the equivalent elevation angle is

$$\vartheta = \theta - \beta. \tag{24}$$

For range direction, the projection illustration is shown in Fig. 2, where the red line denotes the outline of parabolic antenna (in side view) and the blue line is the projection result. Considering the relationship between elevation angle of LOS and pitch angle of attitude, the equivalent elevation angle in (24) can be either positive or negative. However, the intercept R_2 in range direction is always positive. In this vein, an absolute operation is needed in the relationship between R_2 and equivalent elevation angle as

$$R_2 = r|\sin(\theta - \beta)| \tag{25}$$

where $|\cdot|$ is the absolute operation.

For azimuth direction, considering the circle shape of parabolic antenna, the pitch angle of parabolic antenna has no effect on the projection length. Assume that the antenna is parallel with $\boldsymbol{U}\boldsymbol{O}_{\text{Target}}\boldsymbol{V}$ plane (i.e., $\beta = 90°$), the intercept R_1 in azimuth direction can be similarly defined according to (25) as

$$R_1 = r|\sin(\theta - 90°)| = r|-\cos(\theta)| = r|\cos(\theta)|. \tag{26}$$

Note that the projection lengths along the azimuth direction under different observation geometries remain the same as (26), which is independent of the pitch angle in attitude.

Therefore, the projection lengths along the azimuth direction are only related to the elevation angle in LOS. Meanwhile, the length in range direction is related to the equivalent elevation angle, which contains the pitch angle information of target. Following this rational, the radius and pitch angle parameters can be obtained based on the scaling effect in ISAR images.

For F_2 key frame ISAR images with $\phi = \alpha$, according to (26), the original radius r is obtained by the intercept R_1 in azimuth direction and elevation angle θ as

$$r = \frac{1}{F_2}\sum_{f_2=1}^{F_2}\left(\frac{R_{1f_2}}{|\cos\theta_{f_2}|}\right). \tag{27}$$

3) Pitch Angle Estimation: Third, the pitch angle β of parabolic antenna is analyzed. According to (25), the equivalent elevation angle ϑ is derived by intercept R_2 in range direction and radius r of parabolic antenna. In practice, the projection lengths of target in range direction are always positive. Therefore, for F_2 key frame ISAR images, the equivalent elevation angle can be derived from each frame as

$$|\vartheta| = \frac{1}{F_2}\sum_{f_2=1}^{F_2}\arcsin\left(\frac{R_{2f_2}}{r}\right). \tag{28}$$

According to (24) and (28), the pitch angle β is

$$\beta = \frac{1}{F_2}\sum_{f_2=1}^{F_2}\left(\left\{\theta_f \pm \arcsin\left(\frac{R_{2f_2}}{r}\right)\right\}_{\text{set}}\right) \tag{29}$$

where $\{\cdot\}_{\text{set}}$ denotes the validated solution set of pitch angle. The selection criterion of validation solution is given as follows.

Assume that the frame number is $F_2 (F_2 \geq 2)$, and therefore, $2F_2$ results are available according to (24) and (28). The pitch angle is defined as the mean value of selected F_2 results that have the minimum standard deviation. The detailed procedures are listed in Table I.

TABLE I
PROCEDURE OF PITCH ANGLE PARAMETER ESTIMATION

Procedure of Elevation Parameter Estimation
1.Sort the $2F_2$ results in ascending order.
2.Set $i = 1$.
3.Select F_2 sequential results with start serial i, labeled as set $\{F^i\}$.
4.Calculate the standard deviation σ_i and mean value υ_i of set $\{F^i\}$.
5. $i = i + 1$.
6.if $i \leq F_2 + 1$, move to step 3; else, move to step 7.
7.Find the serial p which minimums σ_i
8. $\beta = \upsilon_p$.

4) Position Parameter Estimation: Finally, for position parameters, the estimation process can be conducted based on the projection relationship in (10). Three variables need to be estimated for position parameters. Therefore, arbitrary three-frame ISAR images under different observation conditions are sufficient for position parameter estimation. In this work, all the frames adopted for size and attitude parameter estimation are used for position parameter estimation. Take F frames for example ($F = F_1 + F_2$), F sets of ellipse parameters are available, labeled as

$$\mathbb{R}_f = \begin{bmatrix} x_{0f} & y_{0f} & R_{1f} & R_{2f} & \varphi_f \end{bmatrix}, \quad f = 1, 2, \ldots, F. \tag{30}$$

According to the projection relationship, the centers of ellipses in ISAR images are the projection results of parabolic antenna centers. In this vein, the loss function is constructed as

$$J(\mu, \nu, \omega) = \sum_{f=1}^{F}\left\|\mathbf{P}' \cdot \begin{bmatrix} \mu & \nu & \omega \end{bmatrix}^{\mathrm{T}} - \begin{bmatrix} x_{0f} & y_{0f} \end{bmatrix}^{\mathrm{T}}\right\|_2 \tag{31}$$

where $\mathbf{P}'$ means the projection matrix and $\|\cdot\|_2$ is two-norm.

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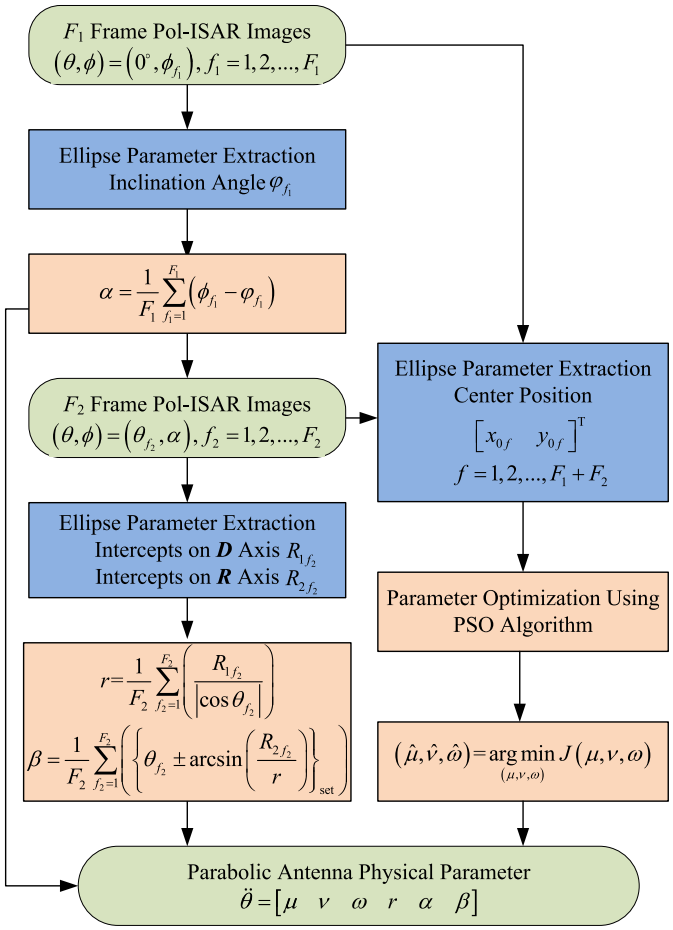


Fig. 3. Flowchart of the proposed method.

Particle swarm optimization (PSO) method [31], [32], which is widely adopted in parameter optimization, is used in this work. The position parameters of parabolic antenna center are obtained by

$$(\hat{\mu},\hat{\nu},\hat{\omega})=\arg\min_{(\mu,\nu,\omega)} J(\mu,\nu,\omega). \tag{32}$$

To sum up, all the six parameters of parabolic antenna can be estimated. The flowchart of the proposed method is shown in Fig. 3.

First, various polarimetric features are derived from Pol-ISAR images, which are adopted for ellipse structure extraction and joint physical parameter estimation.

Then, F_1 frame Pol-ISAR images with elevation angle $\theta=0^\circ$ are adopted for yaw angle α estimation, where $F_1\geq 1$.

Next, F_2 frame Pol-ISAR images with azimuth angle $\phi=\alpha$ are chosen to derive size parameter r and pitch angle β, where $F_2\geq 2$.

Finally, all the F frame Pol-ISAR images are used for the center position $[\mu\ \nu\ \omega]^{\mathrm{T}}$ estimation using the PSO method, where $F\geq 3$.

In Fig. 3, the green shapes indicate the input and output of algorithm. Blue shapes mean the ellipse parameter extraction procedures. Orange shapes denote the parameter estimation processes. Note that in extreme case, the minimum frame number is $F=3$ with $F_1=1$ and $F_2=2$.

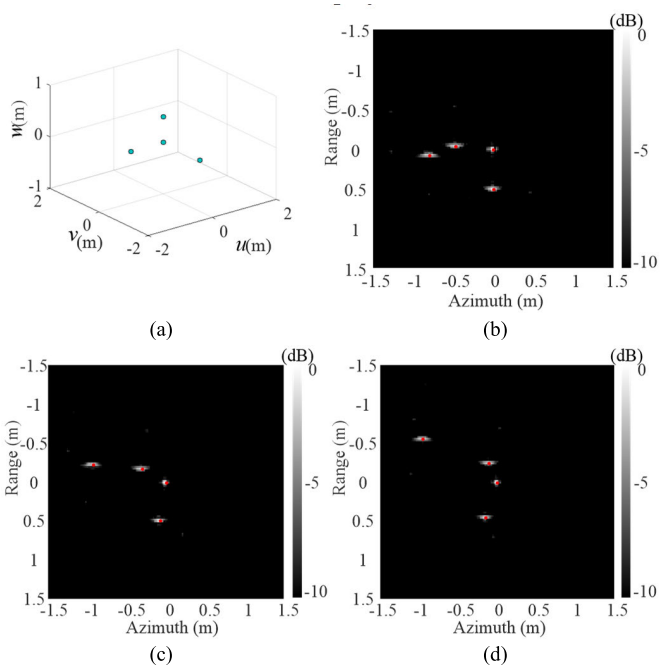


Fig. 4. Strong scatters validation of projection relationship. (a) Original coordinates. (b)–(d) Imaging results of $(\theta,\phi)=(45^\circ,-20^\circ)$, $(\theta,\phi)=(45^\circ,0^\circ)$, and $(\theta,\phi)=(45^\circ,20^\circ)$, respectively.

III. Validation on Point Target Simulation Data

A. Projection Relationship Validation

First, the projection relationship from 3-D space to 2-D ISAR imaging plane is validated. In this part, four strong targets are adopted, the coordinates of which are (0, 0, 0), (−1, 0, 0), (0, 0, 0.5), and (0, −1.5, 0). Moreover, to validate the correctness of rotation matrices in (6), the scatter points are rotated along $\boldsymbol{O}_{\text{Target}}$-$\boldsymbol{UVW}$ with $[\theta_\mu\theta_\nu\theta_\omega]^{\mathrm{T}}=[30^\circ\ 30^\circ\ 30^\circ]^{\mathrm{T}}$.

The original strong scatters in $\boldsymbol{O}_{\text{Target}}$-$\boldsymbol{UVW}$ are shown in Fig. 4(a). The calculated projection results of strong scattering points under different observation conditions can be obtained using (10). The imaging results of simulation data are shown in Fig. 4(b)–(d), where the calculated results are marked with red points. It can be seen that the calculated results are consistent with simulation results. In this vein, the correctness and effectiveness of projection matrix are validated.

Then, point target simulation data of parabolic antenna is adopted for validation. Considering the scattering characteristics of man-made target in ISAR images, only edge points of parabolic antenna are simulated, as shown in Fig. 5. Red points indicate the edge points of parabolic antenna. Green point $\boldsymbol{O}_{\text{Antenna}}$ is the antenna center, and in this case, $\boldsymbol{O}_{\text{Antenna}}=[0.3,\ 0.5,\ 1]^{\mathrm{T}}$. $\mathbf{k}$ is the normal vector of antenna, which starts from $\boldsymbol{O}_{\text{Antenna}}$ and points to the outer space. $\mathbf{k}'$ and $\mathbf{k}''$ are the projection results of $\mathbf{k}$ on the $\boldsymbol{UO}_{\text{Target}}\boldsymbol{W}$ plane and $\boldsymbol{UO}_{\text{Target}}\boldsymbol{V}$ plane. The yaw angle is $\alpha=5^\circ$ and the pitch angle is $\beta=70^\circ$. The radius is $r=1.65$ m.

B. Yaw Angle Estimation

The attitude information of parabolic antenna in 3-D space is denoted by normal vector $\mathbf{k}$. Two attitude parameters that need to be estimated are yaw angle α and pitch angle β.

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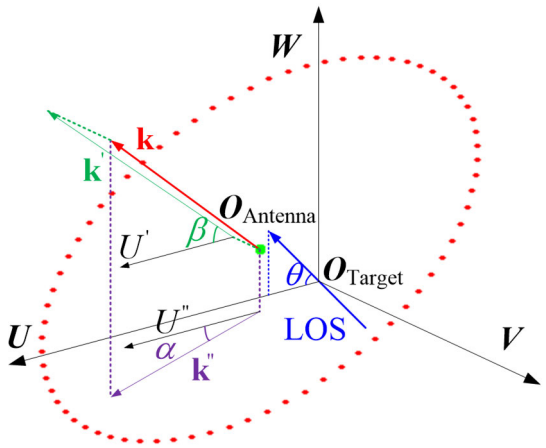


Fig. 5. Imaging geometry of parabolic antenna target.

TABLE II
YAW ANGLE ESTIMATION OF DIFFERENT FRAMES

Frame	1	2	3	4
φ	–5°	0°	5°	10°
ϕ	0°	5°	10°	15°
$\alpha = \phi - \varphi$	5°	5°	5°	5°

For yaw angle α, it can be derived by inclination angle of ellipse using (23) when the elevation angle in LOS is fixed as $\theta = 0°$. In this section, four frame ISAR images are adopted, whose elevation angles are all set as 0°, while azimuth angles are randomly selected as 0°, 5°, 10°, and 15°. The corresponding imaging results are shown in Fig. 6.

The ellipse inclination angles φ, azimuth angles ϕ in LOS, and the derived yaw angles α are listed in Table II. The final result derived from four frames is $\alpha = 5°$, which is equal to the ground truth in Section III-A.

C. Size Parameter Estimation

As mentioned in Section II-D, radius estimation requires the specific observation of $\psi = 0°$.

The point target model in Section III-A is also adopted in this part. Based on the estimation results of yaw angle in Section III-B, the azimuth angle of LOS parameter is defined as $\phi = \alpha - \psi = 5°$ to meet the requirement of radius estimation. The elevation angle θ is randomly selected as 30°, 40°, 50°, and 60°. The corresponding imaging results are shown in Fig. 7. It can be seen that the projection results are all standard ellipses. The intercepts R_1 and R_2 are summarized in Table III.

Based on (27), the radius of parabolic antenna can be obtained as 1.65 m, which is equal to the ground truth in Section III-B.

D. Pitch Angle Estimation

For pitch angle β, it can be derived by (29) and the procedure in Table I. Take imaging results in Section III-C for instance, the $2F$ estimation results of pitch angles are listed in Table IV. The pitch angle results in ascending order are $[-10°, 10°, 30°, 50°, 70°, 70°, 70°, 70°]$.

Apparently, the last four results have the minimum standard deviation value. Therefore, their mean value is the final result of pitch angle, which is $\beta = 70°$. The estimation result of pitch angle is equal to the ground truth in Section III-A.

TABLE III
RADIUS PARAMETER OF DIFFERENT FRAMES

Frame	1	2	3	4
R_1 (m)	1.43	1.26	1.06	0.83
R_2 (m)	1.06	0.82	0.56	0.29
$\lvert\cos\theta\rvert$	0.87	0.77	0.64	0.50
$r = R_1 / \lvert\cos\theta\rvert$ (m)	1.65	1.65	1.65	1.65

TABLE IV
PITCH ANGLE ESTIMATION OF DIFFERENT FRAMES

Frame	1	2	3	4
$\beta = \theta - \arcsin(R_{2f} / r)$	–10°	10°	30°	50°
$\beta = \theta + \arcsin(R_{2f} / r)$	70°	70°	70°	70°

TABLE V
ELLIPSE CENTERS OF DIFFERENT FRAMES

Frame	1	2	3
Range (m)	1.01	1.11	1.15
Azimuth (m)	0.36	0.21	0.03

E. Position Parameter Estimation

According to the optimization functions of (32), more than three ISAR images are required for position parameter estimation. Note that the optimization algorithm needs no specific observation conditions. Therefore, any three different frames are sufficient for position parameter estimation.

Take the imaging geometry in Section III-A for instance, the imaging results of three randomly selected observation LOSs are shown in Fig. 8, whose elevation angles are all 50° and azimuth angles are randomly selected as 0°, 20°, and 40°. The coordinates of ellipse centers in ISAR images under different LOSs are recorded in Table V.

According to (32), the position parameters of parabolic antenna center can be derived using optimization method, such as PSO [31], [32]. Parameter optimization experiments are repeated ten times. The mean values of estimation results are $[0.3\ 0.5\ 1]^{\mathrm{T}}$, which are the same as the ground truth in Section III-A. Parameter estimation experiments under other conditions (e.g., various elevation and azimuth angles) are also conducted, and the position parameters can be correctly estimated as well.

F. Physical Parameter Joint Estimation

From Sections III-B to III-E, the parabolic antenna parameters are estimated individually. In this part, all these six physical parameters of parabolic antenna are jointly estimated using the same set of ISAR frames.

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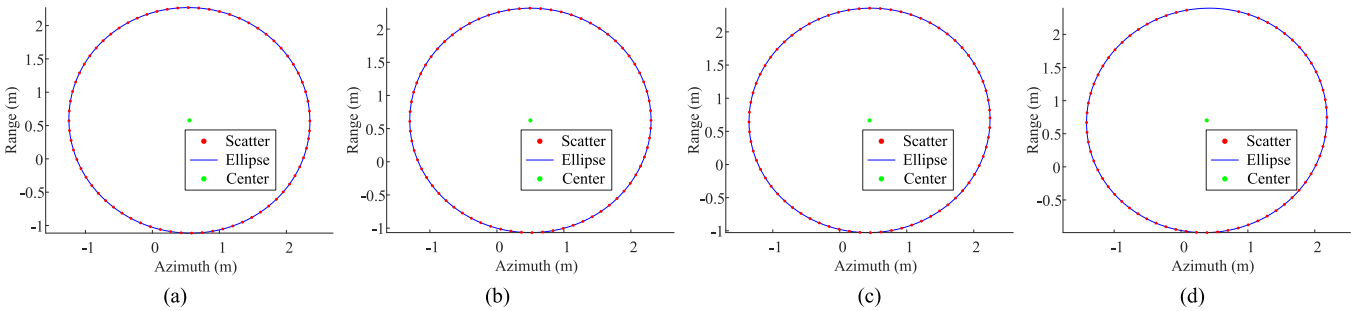


Fig. 6. Imaging results under different LOS azimuth angles. (a) $\phi = 0°$ and $\theta = 0°$. (b) $\phi = 5°$ and $\theta = 0°$. (c) $\phi = 10°$ and $\theta = 0°$. (d) $\phi = 15°$ and $\theta = 0°$.

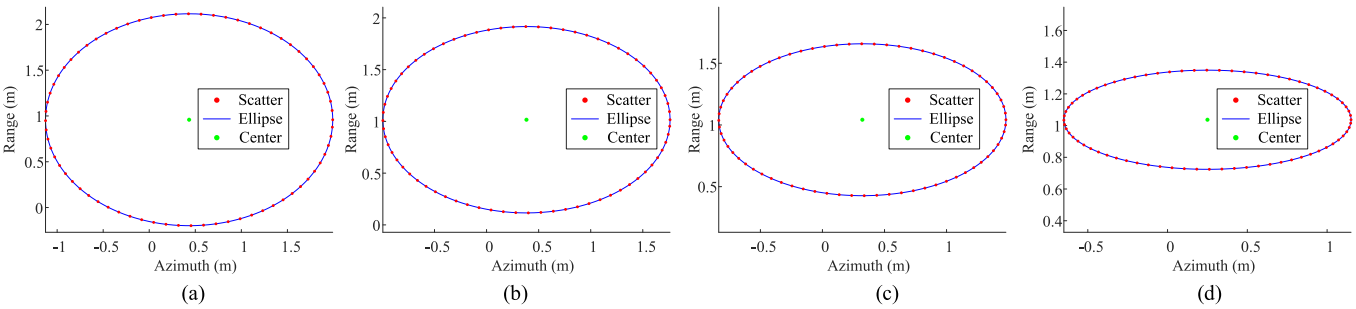


Fig. 7. Imaging results under different LOS elevation angles. (a) $\phi = 5°$ and $\theta = 30°$. (b) $\phi = 5°$ and $\theta = 40°$. (c) $\phi = 5°$ and $\theta = 50°$. (d) $\phi = 5°$ and $\theta = 60°$.

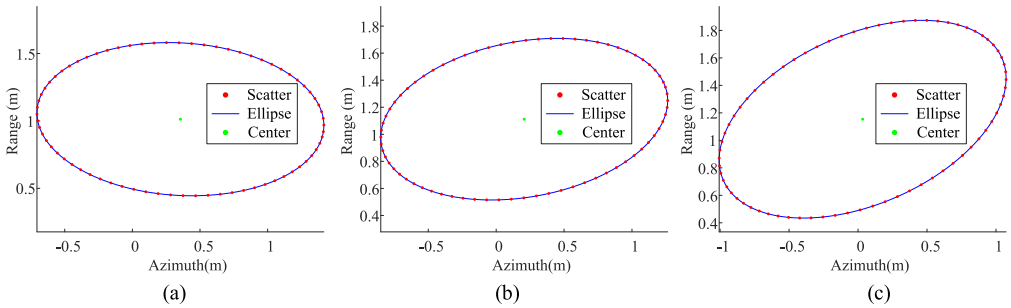


Fig. 8. Imaging results under different LOS azimuth angles. (a) $\phi = 0°$ and $\theta = 50°$. (b) $\phi = 20°$ and $\theta = 50°$. (c) $\phi = 40°$ and $\theta = 50°$.

TABLE VI
LOS OF THREE FRAMES REQUIRED

Frame	1	2	3
ψ	ψ_1	0°	0°
θ	0°	θ_1	θ_2

In the extreme case, three key frame ISAR images are sufficient for six physical parameters estimation of parabolic antenna. The corresponding observation angles are listed in Table VI, where $\psi_1 \neq 0$ and $\theta_1 \neq \theta_2$. Note that the elevation angle in frame 1 should be 0° and the equivalent azimuth angles ψ in frames 2 and 3 should be 0°.

In this part, a parabolic antenna with different parameters is adopted. The radius of antenna is $r = 1.8$ m. The coordinate of $\boldsymbol{O}_{\text{Antenna}}$ is $[0.2\ -0.8\ 1.5]^{\text{T}}$. The yaw and pitch angles of antenna are $\alpha = 20°$ and $\beta = 60°$.

To start with, the LOS of frame 1 is determined with a randomly selected azimuth angle $\phi = 25°$. Thus, the LOS of frame 1 is $(\phi, \theta) = (25°, 0°)$. The LOS parameters of frames 2 and 3 are determined thereafter.

1) Yaw Angle Estimation: For frame 1, the extracted ellipse parameters are summarized in Table VII. The inclination angle is $\varphi = 5°$. According to (23), the yaw angle can be estimated with only one frame, which is $\alpha = \phi - \varphi = 20°$.

The azimuth angles of frames 2 and 3 can be determined as $\phi_2 = \phi_3 = \alpha = 20°$ to ensure that $\psi_2 = \psi_3 = 0°$. The corresponding elevation angles of frames 2 and 3 are randomly selected as $\theta_2 = 10°$ and $\theta_3 = 0°$. In this vein, the LOS parameters are $(\phi, \theta) = (25°, 0°)$, $(\phi, \theta) = (20°, 10°)$, and $(\phi, \theta) = (20°, 0°)$ for frames 1–3, respectively. The imaging results of three frames are shown in Fig. 9, while the ellipse fitting results are summarized in Table VII.

2) Size Parameter Estimation: Frames 2 and 3 are adopted for size parameter estimation. Intercepts R_1 in two frames are 1.77 and 1.80. The radius parameter is obtained using (27), which is $r = 1.80$ m.

3) Pitch Angle Estimation: The pitch angle β is estimated based on frames 2 and 3. Intercepts R_2 are 1.38 and 1.56.

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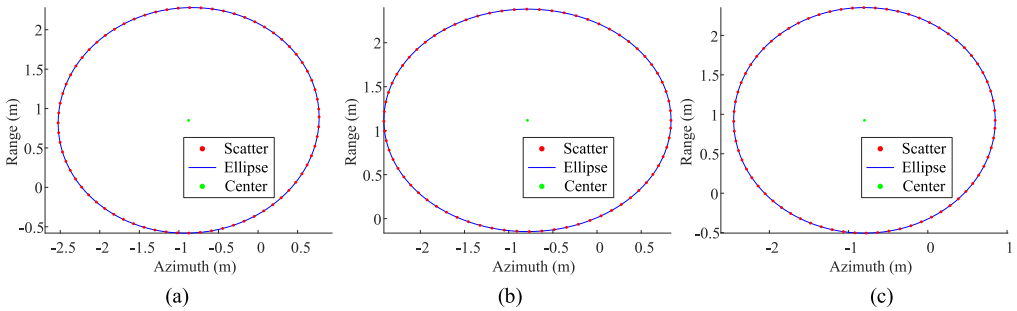


Fig. 9. Imaging results of point targets under different LOSs. (a) $\phi = 25°$ and $\theta = 0°$. (b) $\phi = 20°$ and $\theta = 10°$. (c) $\phi = 20°$ and $\theta = 0°$.

TABLE VII
ELLIPSE PARAMETERS OF TARGET UNDER DIFFERENT OBSERVATION LOS PARAMETERS

No.	LOS	x_0 (m)	y_0 (m)	R_1 (m)	R_2 (m)	φ (°)
1	$(\phi,\theta)=(25°,0°)$	–0.88	0.85	1.80	1.56	5.00
2	$(\phi,\theta)=(20°,10°)$	–0.79	1.12	1.77	1.38	0.00
3	$(\phi,\theta)=(20°,0°)$	–0.80	0.92	1.80	1.56	0.00

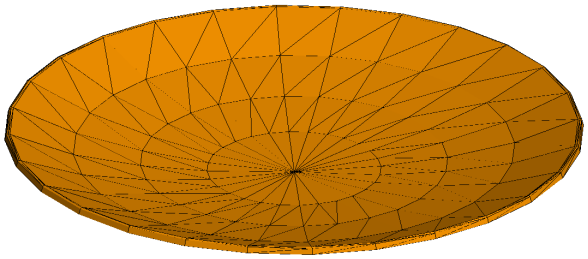


Fig. 10. CAD model of parabolic antenna component.

When $r = 1.80$ m, a set of β can be obtained using (29), which are $[-60°, -40°, 60°, 60°]^{\mathrm{T}}$. Apparently, the mean value of last two estimated results is the final pitch angle result, which is $\beta = 60°$.

4) Position Parameter Estimation: Coordinates of parabolic antenna centers are estimated based on ellipse center parameters of three frames using (31) and (32). Experiments are repeated ten times. The mean optimization results are $\boldsymbol{O}_{\mathbf{Antenna}} = [0.20, -0.80, 1.50]^{\mathrm{T}}$.

In summary, the estimation results of these six parameters are consistent with the ground truth. In the extreme case, three key frame ISAR images are sufficient for parameter estimation. Experimental studies with parabolic antenna point target simulated data validate the accuracy of the joint parameter estimation method.

IV. EXPERIMENTAL STUDIES

In this section, the proposed parameter estimation method is validated on electromagnetic simulation data of parabolic antenna component and complex satellite of QPS-SAR (A satellite developed by Institute for Q-Shu Pioneers of Space, Japan). The electromagnetic simulation parameters are listed in Table VIII.

The extraction of ellipse structure is the primary process of the proposed method. Therefore, the parameters of parabolic antenna (e.g., radius and position parameters) in real satellite can be arbitrary as along as the imaging results of antenna (i.e., ellipse structure) are large enough to be extracted. Besides, the imaging quality of ISAR image is assumed to be good enough to extract ellipse structures.

A. Experiments With Parabolic Antenna Model

The CAD model of parabolic antenna is shown in Fig. 10. The radius size is $r = 1.6233$ m. The position of antenna center is $[0\ 0\ 0.3]^{\mathrm{T}}$. The parabolic antenna is placed horizontally, with pitch angle $\beta = 90°$ and arbitrary yaw angle α.

Note that the yaw angle of parabolic antenna is estimated by inclination angle of ellipse. Since the yaw angle is arbitrary, the equivalent azimuth angle remains 0° under all observation conditions. Therefore, the projection results of parabolic antenna are all standard ellipse, whose inclination angles should be $\varphi = 0°$. According to (23), the estimation error of inclination angle is the same as yaw angle error when the azimuth angle in LOS is fixed. Thus, it is reasonable to consider the inclination angle error to evaluate the parameter estimation performance when the yaw angle is arbitrary.

In this section, a number of Pol-ISAR images are used for parameter estimation to eliminate the impact of randomly selected frames. Since the yaw angle is arbitrary, the requirement of azimuth angle in the LOS parameter is always satisfied. Therefore, the elevation angle in LOS parameter is the only factor that needs to be considered. According to Section II, at least three frames are needed for position parameters estimation. In this scenario, a total of 90 sets of key frames are collected for parameter estimation. Each set contains three frame Pol-ISAR images with the same azimuth angle and different elevation angles of 50°, 60°, and 70°. The azimuth angles of different sets are 1°∼90° incremented by 1°.

Among the polarimetric features considered in this work, four features perform well in ellipses extraction and subsequent physical parameters estimation, which are single-polarization channel data HV, Yamaguchi decomposition component f_v, general polarimetric decomposition component p_v, and Krogager decomposition component k_h.

Taking $\phi = 1°$ for instance, the ellipse extraction results of four features under different elevation angles are shown in Fig. 11. Red points indicate the extracted strong scatters,

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TABLE VIII
SIMULATION PARAMETERS

Parameters	Values
Central Frequency	16GHz
Polarization	HH, HV, VH, VV
Bandwidth	2GHz
Frequency Step	0.025GHz
Azimuth Angle Range	0°~180°
Azimuth Angle Step	0.1°
Elevation Angle Range	10°~90°
Elevation Angle Step	10°
Resolution (Azimuth×Range)	0.075m×0.075m

TABLE IX
PARAMETER ESTIMATION ERRORS OF PARABOLIC ANTENNA USING DIFFERENT POLARIMETRIC FEATURES

Feature	μ (m)	ν (m)	ω (m)	r (m)	β (°)	φ (°)
HV	–0.0139	0.0050	–0.0080	0.0126	**1.1154**	–0.9869
f_v	–0.0981	–0.1012	0.0547	**0.0036**	1.5777	–1.0832
p_v	0.0532	0.0344	–0.0261	0.0359	1.6959	1.5922
k_h	**–0.0095**	**0.0037**	**–0.0079**	0.0125	1.1396	**–0.0584**

while the blue line denotes the fitting results of ellipses. The centers of ellipses are marked with green points. All these four polarimetric features are effective in extracting ellipses. Since the elevation angle of LOS is 0° and the pitch angle of parabolic antenna is 90°, the projection results of parabolic antennas should be circles under different azimuth angles. From visual inspection, the extraction results from k_h feature are more like circles.

Quantitatively, 90 sets of experimental studies are conducted with different azimuth angles of $\phi = 1°, 2°, \ldots, 90°$, and the mean estimation errors under different observation conditions are listed in Table IX, including position parameter, radius parameter, pitch angle, and inclination angles.

In Table IX, the errors of position parameters are quite small. The maximum position error of HV polarization data is -0.0139 m in the $\boldsymbol{U}$-axis, while that of f_v feature is -0.1012 m in the $\boldsymbol{V}$-axis and that of p_v feature is 0.0532 m in the $\boldsymbol{U}$-axis. The k_h feature has the best performance in position parameter estimation and the estimation errors are less than 0.01 m in all three axes. As for radius parameter, the f_v feature achieves the best performance with 0.0036-m error. The k_h feature has the second least error of 0.0125 m for radius parameter. For attitude parameters, HV data has the least elevation angle error of 1.1154°, while that of k_h feature is the second least with 1.1396°. Moreover, the k_h feature has the best performance in inclination angle estimation with $-0.0584°$ error. Considering all the estimation errors of six parameters, the comprehensive performance of k_h feature is the best among all features, which is suitable for physical parameter joint estimation of parabolic antenna.

Furthermore, the superiorities of selected four polarimetric features are analyzed in detail. In [22], three expansion matrices are adopted to model the volume scattering component as

$$\mathbf{T}_{\text{volume}} = \left\{ \frac{1}{30}\begin{bmatrix} 15 & 5 & 0 \\ 5 & 7 & 0 \\ 0 & 0 & 8 \end{bmatrix} \quad \frac{1}{4}\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \frac{1}{30}\begin{bmatrix} 15 & -5 & 0 \\ -5 & 7 & 0 \\ 0 & 0 & 8 \end{bmatrix} \right\}. \tag{33}$$

For each pixel, a volume scattering matrix is selected based on the magnitude balance between co-polarizations [22].

In [23], apart from the abovementioned three models in (33), another representative volume scattering model is used

$$\mathbf{T}_{\text{volume}} = \frac{1}{3}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{34}$$

In Krogager decomposition scheme [24], helix scattering is defined as

$$\begin{cases} k_h = \text{abs}(|s_{\text{RR}}| - |s_{\text{LL}}|) \\ s_{\text{RR}} = \text{j}s_{\text{HV}} + \dfrac{1}{2}(s_{\text{HH}} - s_{\text{VV}}) \\ s_{\text{LL}} = \text{j}s_{\text{HV}} - \dfrac{1}{2}(s_{\text{HH}} - s_{\text{VV}}). \end{cases} \tag{35}$$

For full-polarimetric data, the magnitude ratio of co-polarization and cross-polarization data can be as high as 40 dB, which means that information contained in cross-polarized data (e.g., HV) is merged with the high-valued co-polarization data (e.g., HH and VV). Researchers have proven that only a small arc of curved lines can be seen in ISAR imaging results [33]. Take parabolic antenna for example, the imaging result of HH polarization data under LOS parameter of $(\phi, \theta) = (0°, 70°)$ is shown in Fig. 12. Only a small arc of parabolic antenna is visible, leading to the failure of ellipse extraction and physical parameter estimation. It should be noted that the same is true for VV polarization data and the SPAN feature.

Moreover, the odd-bounce scattering (i.e., surface scattering and sphere scattering) and double-bounce scattering (i.e., dihedral scattering) are derived from linear combinations of co-polarization data in the complex-valued domain. Therefore, the imaging results of these polarimetric features are similar to Fig. 12, which are unsuitable for ellipse detection and physical parameter estimation. On the contrary, the cross-polarization data are more sensitive to corners and edges [34]. The imaging results of HV polarization data under the same observation condition are shown in Fig. 11(a1). The edge information of parabolic antenna is well reserved in HV polarization data, which is quite suitable for joint parameter estimation.

As for volume scattering, the cross-polarization component is paid special attention to. According to (33) and (34), the ratio of co-polarization [i.e., $\mathbf{T}_{\text{volume}}(1, 1)$ and $\mathbf{T}_{\text{volume}}(2, 2)$] and cross-polarization [i.e., $\mathbf{T}_{\text{volume}}(3, 3)$] components is smaller than 3 dB. In this scenario, the edge information

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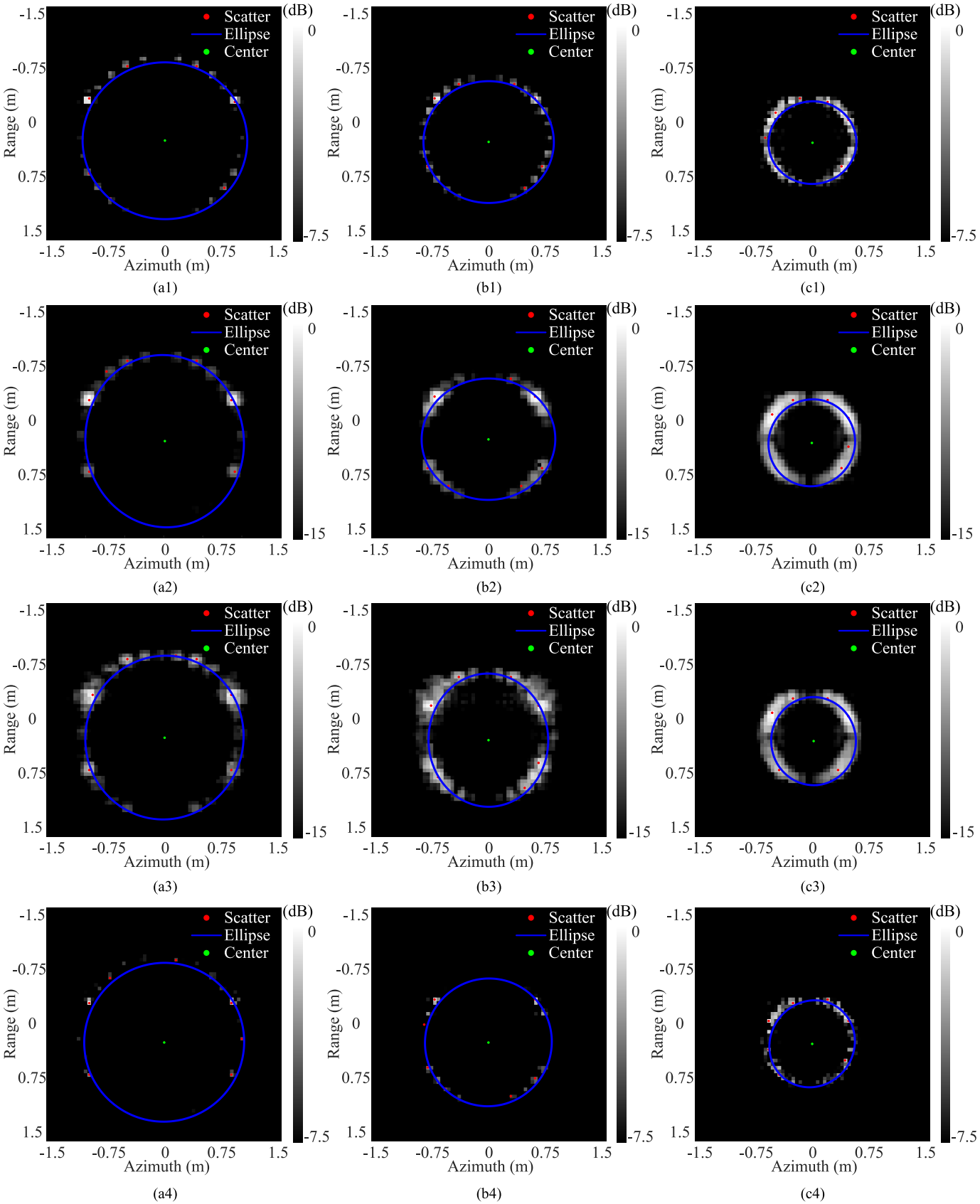


Fig. 11. Polarimetric features of CAD models. (a)–(c) LOS parameters of $(\phi, \theta) = (0°, 50°)$, $(\phi, \theta) = (0°, 60°)$, and$(\phi, \theta) = (0°, 70°)$, respectively. (1)–(4) HV, f_v, p_v, and k_h, respectively.

in cross-polarization data can be revealed, as shown in the second and third rows of Fig. 11. For helix scattering, the co-polarization data are operated in absolute value domain rather than complex value domain, where the detailed information in co-polarization data can also be revealed, as shown in the fourth row of Fig. 11.

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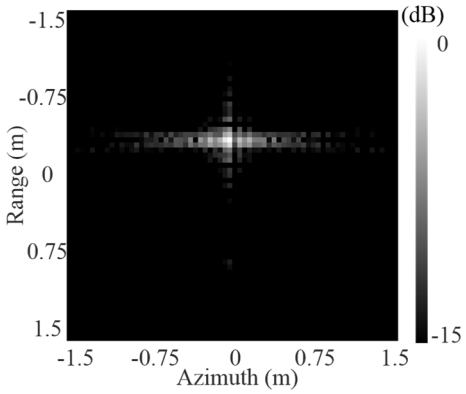


Fig. 12. Imaging result of HH polarization data under LOS $(\phi, \theta) = (0°, 70°)$.

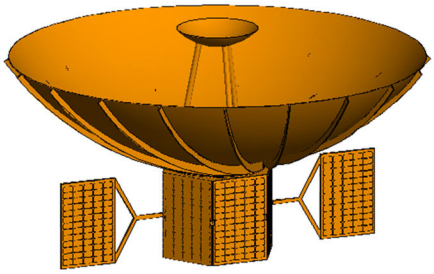


Fig. 13. CAD model of QPS-SAR satellite.

To summarize, among these polarimetric features, HV polarization data, volume scattering component, and helix scattering components are more suitable for ellipse detection and physical parameter estimation. Quantitatively, helix scattering component of Krogager decomposition achieves the best performance for joint parameter estimation.

B. Experiments With QPS-SAR Satellite Model

Apart from separated parabolic antenna component, complex satellite with parabolic antenna is also used for experimental studies. In this part, QPS-SAR model is adopted, the CAD model of which is shown in Fig. 13. QPS-SAR mainly consists of parabolic antenna, satellite cabin, and solar panels. The size of parabolic antenna is 1.8 m, and the center position is $[0\ 0\ 0.9]^{\mathrm{T}}$. The pitch angle $\beta = 90°$and yaw angle α is also arbitrary.

Three frame Pol-ISAR images, whose LOS parameters are $(\phi, \theta) = (45°, 50°)$, $(\phi, \theta) = (45°, 45°)$, and$(\phi, \theta) = (45°, 40°)$, are adopted for joint physical parameter estimation of parabolic antenna. According to Section IV-A, the Krogager decomposition component k_h, which has the best performance for parameter estimation, is adopted in this part. Polarimetric features under different observation conditions are shown in Fig. 14. Red points are the point cloud projection results on ISAR images. The blue line denotes the fitting results of ellipses, while the green points are the centers of ellipses.

The extracted ellipse parameters are summarized in Table X and the parameter estimation results are listed in Table XI. The absolute errors of position parameters and size parameter are all 0, while that of pitch angle is 0.03°, which is rather close to ground truth. In this vein, the effectiveness of the proposed method can be further validated.

TABLE X
ELLIPSE PARAMETERS OF TARGET UNDER DIFFERENT OBSERVATION LOS PARAMETERS

No.	LOS	x_0 (m)	y_0 (m)	R_1 (m)	R_2 (m)	φ (°)
1	$(\phi,\theta)=(45°,50°)$	0.00	0.69	1.16	1.16	0.00
2	$(\phi,\theta)=(45°,45°)$	0.00	0.63	1.27	1.27	0.00
3	$(\phi,\theta)=(45°,40°)$	0.00	0.58	1.38	1.38	0.00

TABLE XI
PARAMETER ESTIMATION RESULTS WITH OPTIMIZATION APPROACH

Parameter	Position	Size (m)	Pitch Angle(°)
Ground Truth	$[0, 0, 0.9]^{\mathrm{T}}$	1.8	0
Estimation Result	$[0.00, 0.00, 0.90]^{\mathrm{T}}$	1.80	0.03
Absolute Error	$[0.00, 0.00, 0.00]^{\mathrm{T}}$	0.00	0.03

V. DISCUSSION

A. Effects of LOS Parameter Errors

Key frame ISAR images acquired under specific observation geometries are required for the proposed method. Nonetheless, in practice, mismatches in observation conditions may impact the accuracy of parameter estimation, the effects of which are analyzed in this part. Note that the "error" here refers to the mismatches between real observation parameters and the required observation parameters.

Theoretically, errors in observation conditions directly affect the extraction results of ellipse parameters, which subsequently affect the parameter estimation results of parabolic antenna. However, the error transmission mechanism cannot be fully characterized by explicit function since the ellipse extraction process does not follow a closed-form equation. To mitigate this issue, numerical sensitivity analysis is conducted using point target simulation data. Note that three position parameters are estimated using optimization algorithms and no specific observation conditions are needed. Therefore, sensitivity analysis of position parameter estimation is omitted here.

According to Section II-D, $\theta = 0°$ is required for yaw angle estimation and $\phi = \alpha$ is required for radius and pitch angle estimation. Therefore, the effects of elevation angle error on yaw angle estimation and effects of azimuth angle error on radius and pitch angle estimation are considered. Point target simulation data are adopted. The radius of antenna r is 1.8 m. The coordinate of $\boldsymbol{O}_{\mathrm{Antenna}}$ is $[0.2\ -0.8\ 1.5]^{\mathrm{T}}$. The yaw angle is randomly fixed as $\alpha = 20°$. Considering the real observation geometry of ISAR systems, typical pitch angles are set as $\beta = 10°$, 30°, 50°, and 70°.

First, the effects of elevation angle error on yaw angle estimation are considered, as shown in Fig. 15(a). Blue triangle, red circle, yellow diamond, and purple asterisk denote the error curves of $\beta = 10°$, 30°, 50°, and 70°, respectively. With the increase of pitch angle, the maximum error of yaw angle estimation is getting larger. Besides, the symmetry axes of different error curves are closer to 0° as the pitch angle increases. For each error curve, yaw angle error is sensitive

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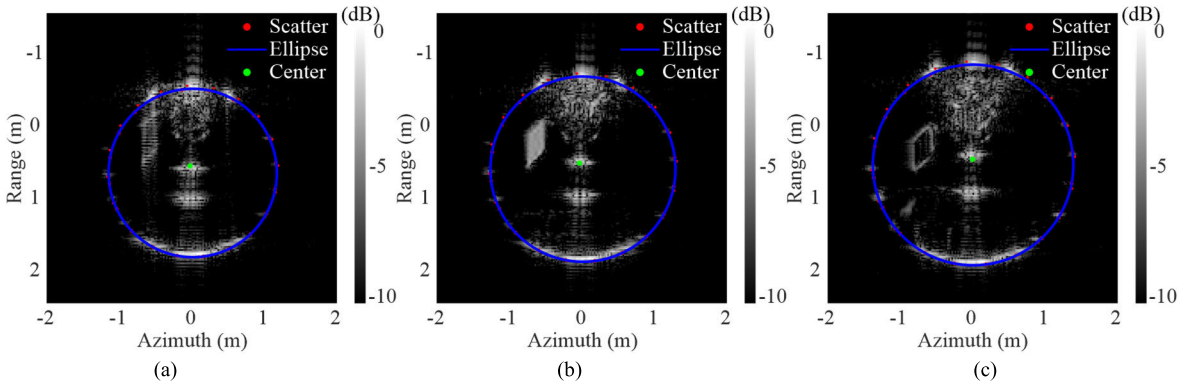


Fig. 14. Polarimetric features of QPS-SAR target in different LOSs. (a) $(\phi, \theta) = (45°, 50°)$. (b) $(\phi, \theta) = (45°, 45°)$. (c) $(\phi, \theta) = (45°, 40°)$.

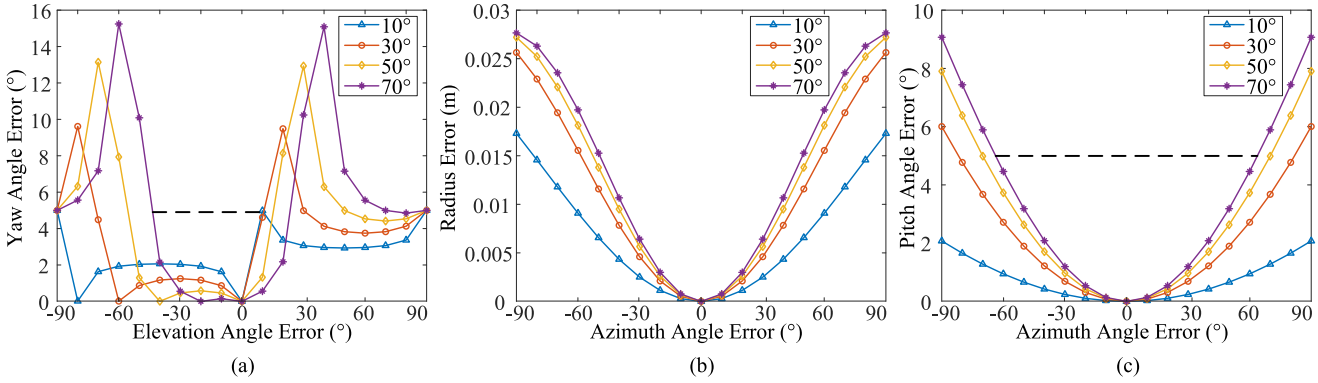


Fig. 15. Effects of observation angle error on parameter estimation. (a) Elevation angle error on yaw angle estimation. (b) Azimuth angle error on radius estimation. (c) Azimuth angle error on pitch angle estimation.

to elevation angle errors. Take all the different elevation angle situations into consideration, if the yaw angle error is supposed to be smaller than 5°, the corresponding elevation angle error ranges are [−43.59°, 10.81°], as shown in Fig. 15(a) with black dashed line.

Next, the effects of azimuth angle error on parameter estimation are analyzed, which are shown in Fig. 15(b) and (c). Blue triangle, red circle, yellow diamond, and purple asterisk denote the error curves of $\beta = 10°$, 30°, 50°, and 70°, respectively. As the pitch angle increases, the maximum errors of radius and pitch angle estimation results are both getting larger. The error curves of radius and pitch angle errors are all symmetric with respect to 0°. When $\beta = 70°$, the maximum radius error is 0.0277 m, accounting for 0.15% of the original radius parameter. The maximum error of pitch angle estimation results is 9.07°. It can be seen that the radius and pitch angle estimation results are more stable compared with yaw angle estimation results. The reason behind this phenomenon is that the radius and pitch angle are estimated from the lengths of intercepts, which are more robust than the inclination angle parameters. Similarly, for 5° error tolerance, the corresponding azimuth angle errors are supposed to be within the ranges of [−63.85°, 63.85°].

According to Section II, the estimation results of yaw angles are set as the azimuth angles in LOS for radius and pitch angle estimation. In this scenario, the azimuth angle errors in Fig. 15(b) are equal to the yaw angle estimation errors in Fig. 15(a). The maximum error of yaw angle estimation in Fig. 15(a) is 15.24°, which is within the error tolerance [-63.85°, 63.85°]. Therefore, the arbitrary elevation angle in observation conditions can meet 5° error tolerance of radius estimation and 0.15% error tolerance of pitch angle estimation.

Based on the sensitivity analysis of observation angle errors, conclusions can be drawn as follows.

1) The original pitch angle of target has effects on the error tolerance of LOS parameters.

2) The elevation angle error has an obvious impact on yaw angle estimation.

3) The azimuth angle error is inherited from yaw angle estimation error. Therefore, the mismatch of elevation angle also has effects on radius and pitch angle estimation.

4) The estimation results of radius and pitch angle estimation are relatively stable under different azimuth angle errors since the lengths of intercepts are more robust in ellipses.

B. Effects of SNR

The accurate extraction of ellipse structure is the basic procedure of the proposed method, which can be affected by the imaging quality of ISAR images. SNR is one of the key factors in ISAR images, the effects of which are analyzed in this part.

The SNR is defined as

$$\mathrm{SNR} = 10\log_{10}(E_s/E_n) \tag{36}$$

where E_s denotes the signal power in echo domain and E_n is the power of added Gaussian white noise.

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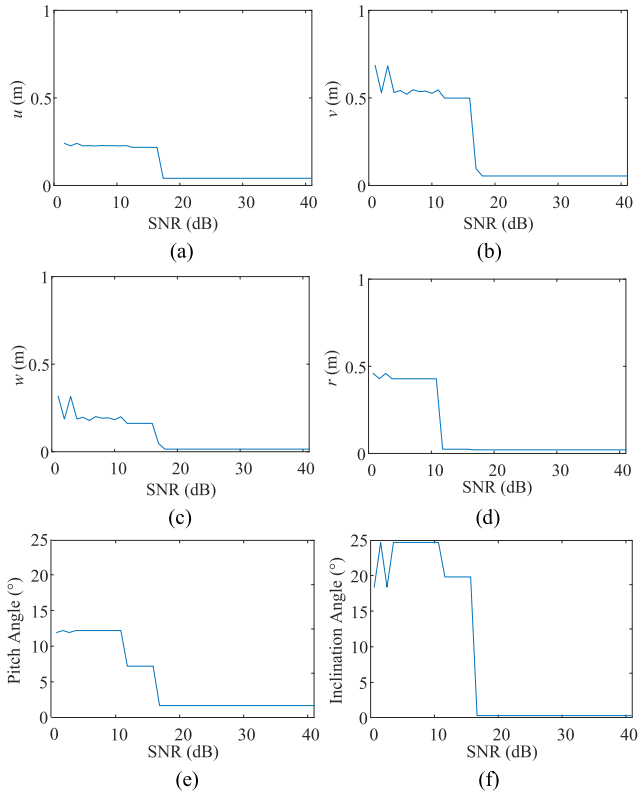


Fig. 16. Effects of SNR on parameter estimation errors. (a)–(c) are coordinates in the $\boldsymbol{U}$-, $\boldsymbol{V}$-, and $\boldsymbol{W}$-axes, respectively. (d) Radius. (e) Pitch angle. (f) Inclination angle.

Take feature k_h for example, the parameter estimation errors under different SNRs are shown in Fig. 16. Generally speaking, the errors of parameter estimation converge with the increase of SNR. In detail, when the SNR is larger than 17 dB, the errors of position parameters converge to 0.01 m and the error of radius parameter is 0.01 m. The estimation errors of pitch angle and inclination angle converge to 1.2° and 0.1°.

In summary, the effects of SNR on parameter estimation are obvious. A similar phenomenon can be seen in other polarimetric features. In practice, 17-dB higher SNR is recommended for parameter estimation.

C. Scenario of Acquiring Key Frames

The projection relationship of parabolic antenna in 3-D space and ellipse structure in 2-D ISAR images holds on the basis of specific observation conditions.

First, the elevation angle of LOS should be 0° for yaw angle estimation. At the beginning and ending of a transit, elevation angles are rather close to 0°, which can satisfy the specific requirement. Second, the azimuth angle of LOS should be equal to the estimated yaw angle. During the whole transit, the range of azimuth angle is quite large, which is reasonable to find the corresponding observation condition. Theoretically, the required key frames can be obtained in practice.

Take QPS-SAR satellite for example, a ground-based radar is located in Beijing (39.93°N, 116.39°E). The relative LOS parameters between radar and satellite are obtained by the satellite tool kit (STK) software. For a typical transit on September 14, 2023, 1:44–1:54 A.M., the elevation and azimuth angles are shown in Fig. 17. During the 600-s transit time, the dynamic ranges of elevation angle are [0°, 33.15°], while those of azimuth angle are [21.02°, 174.95°].

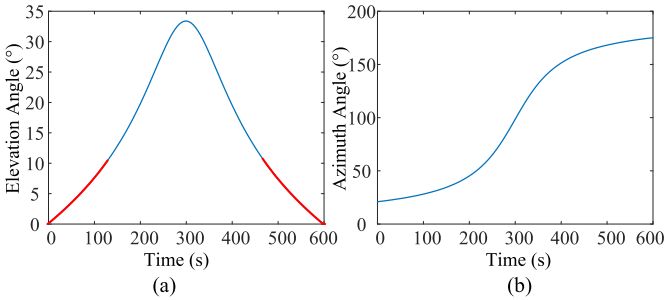


Fig. 17. LOS parameters. (a) Elevation angle. (b) Azimuth angle.

According to Section II-B, $\theta = 0°$ is required for yaw angle estimation, which can be satisfied in practice according to the elevation angle of LOS parameters. Moreover, as discussed in Section V-A, when the error of yaw angle is supposed to be smaller than 5°, the corresponding elevation angle error range is [−43.59°, 10.81°]. In this vein, the effective imaging time is 265 s, accounting for 44.17% of the whole satellite transit. The effective imaging time is marked with red line in Fig. 17(a).

As for radius and pitch angle estimation, the azimuth angle of LOS should be consistent with the estimated yaw angle. The dynamic ranges of azimuth angle in Fig. 17(b) are quite large for radius and pitch angle estimation. Besides, according to Section V-A, for pitch angle parameter estimation with 5° error tolerance, the azimuth angle errors are within [−63.85°, 63.85°]. It is reasonable to believe that the corresponding frames with specific azimuth angle can be found during one satellite transit.

Note that a number of satellites and radar stations' locations have been examined and the characteristics of LOS parameters are as follows: 1) elevation angle is close to 0° at the beginning and ending of transit and 2) the range of azimuth angle is near 180° during the whole transit.

In summary, from the viewpoint of key frame acquiring, the basic requirement of the proposed method can be satisfied in practice.

VI. Conclusion

Accurate estimation of satellite physical parameters is of vital importance for space situation awareness. To address this issue, a physical parameter joint estimation method for satellite parabolic antenna with key frame Pol-ISAR images is developed in this work, where physical parameters include size, attitude, and position parameters.

The direct mapping relationships between parabolic antenna parameters in 3-D space and ellipse structure parameters in 2-D ISAR images are investigated for the first time. Theoretically, three key frame ISAR images are sufficient for parameter estimation in extreme cases. Closed-form expressions are deduced, providing an efficient way for physical parameters estimation. Moreover, polarimetric decomposition features are adopted for ellipse structure extraction and the following physical parameter estimation. The effectiveness of

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the proposed method is validated with point target simulation data and electromagnetic simulation data.

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